Contents

1 Measure Theory

1.1 Outer Measure

Definition 1.1.1 ▸ Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an ***outer measure*** on X if the following conditions are satisfied:

(i) $\varphi(\varnothing) = 0$,

(ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,

(iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,

(iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X.

Example 1.1.2 ▸ Outer Measure

(i) In an arbitrary set X, define $\varphi(A) = 1$ if A is nonempty and $\varphi(\varnothing) = 0$.

(ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A.

(iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$

(iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

(v) Select a fixed x_0 in an arbitrary set X, and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the ***Dirac measure*** concentrated at x_0.

Definition 1.1.3

Let φ be an outer measure on a set X. A set $E \subset X$ is called ***φ-measurable*** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ-measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Note.
也就是说, E 是可测的, 当且仅当任意集合 A 都可以被 E 按外测度被良好地拆解为 $A \cap E$ 和 $A \setminus E$.

Lemma 1.1.4

A set $E \subset X$ is φ-measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Note.
也就是说 E 可测, 当且仅当落在 E 的内, 外的集合在外测度意义下是无关的.

Proof. For Necessity, if E is φ-measurable , then for all such P, Q,

$$\varphi\left(P \cup Q\right) = \varphi\left(\left(P \cup Q\right) \cap E\right) + \varphi\left(\left(P \cup Q\right) - E\right) = \varphi\left(P\right) + \varphi\left(Q\right)$$

For sufficiency , let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi\left(A\right) = \varphi\left(P \cup Q\right) = \varphi\left(P\right) + \varphi\left(Q\right) = \varphi\left(A \cap E\right) + \varphi\left(A \setminus E\right)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X. Then the following four statements hold:

(i) E is φ-measurable whenever $\varphi(E) = 0$.
(ii) $\varnothing$ and X are φ-measurable.
(iii) $E_1 - E_2$ is φ-measurable whenever E_1 and E_2 are φ-measurable.
(iv) If $\{E_i\}$ is a ***countable collection of disjoint φ-measurable sets***, then

$\bigcup_{i=1}^{\infty} E_i$ is φ-measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately frome lemma 1.1.4.
3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implis that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally , we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^{k} E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ-measurable and that

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}
\varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c\right) \\
&= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c \cap S_k^c\right) + \varphi\left(A \cap E_{k+1}^c \cap S_k\right) \\
&= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right) + \varphi(A \cap S_k)
\end{aligned}$$

Replace A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^{k} \varphi(A \cap S_k \cap E_i) + \varphi\left(A \cap S_k \cap S_k^c\right) = \sum_{i=1}^{k} \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

this , inturn , implies that S_{k+1} is φ-measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \le \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

$\square$

Lemma 1.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable, and so is A_i.

Proof. Let $S_k = \bigcup_{i=1}^{k} E_i$. Let $A_k = S_k \setminus S_{k-1}, k \ge 1$, where $S_0 = \varnothing$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem . Inductively , every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable . $\square$

Theorem 1.1.7

If $\{E_i\}$ is a sequence of φ-***measurable*** sets in X, then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ-***measurable***.

Proof. By the lemma above, there exists a sequence of disjoint φ - measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 1.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i$ is as well . Since X is measurable , $E_i^c = X \setminus E_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable . Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable . □

Definition 1.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a σ-***algebra***:

(i) if $E \in \Sigma$, then $E^c \in \Sigma$;

(ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ.

Definition 1.1.9

In a topological space, the elements of the smallest σ-algebra that contains all open sets are called ***Borel sets***. The term "smallest" is taken in the sense of inclusion.

Proof. We need to show that the "smallest" exists. □

Corollary 1.1.10

If φ is an ***outer measure*** on an arbitrary set X, then the class of φ-measurable sets forms a σ-***algebra***.

Proof. Follows immediately from Theorem 1.1.5 and Theorem 1.1.7. □

Corollary 1.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ-measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i, then

$$\varphi(\bigcup_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i, and if $\varphi(E_{i_0}) < \infty$ for some i_0, then

$$\varphi(\bigcap_{i=1}^{\infty} E_i) = \varphi(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ-measurable sets, then

$$\varphi(\liminf_{i\to\infty} E_i) \leq \liminf_{i\to\infty} \varphi(E_i).$$

(vi) If

$$\varphi(\bigcup_{i=i_0}^{\infty} E_i) < \infty$$

for some positive integer i_0, then

$$\varphi(\limsup_{i\to\infty} E_i) \geq \limsup_{i\to\infty} \varphi(E_i).$$

Proof. (v) Let $A_j = \bigcap_{k=j}^{\infty} E_k$, then

$$\liminf_{i\to\infty} E_i = \lim_{j\to\infty} A_j = \bigcup_{j=1}^{\infty} A_j$$

It follows from (iv) that

$$\varphi\left(\liminf_{i\to\infty} E_i\right) = \lim_{j\to\infty} \varphi\left(A_j\right)$$

Since $A_j \subseteq E_j$, we have $\varphi\left(A_j\right) \leq \varphi\left(E_j\right)$, then

$$\lim_{j\to\infty} \varphi\left(A_j\right) \leq \liminf_{i\to\infty} \varphi\left(E_j\right)$$

Thus

$$\varphi\left(\liminf_{i\to\infty} E_i\right) \leq \liminf_{i\to\infty} \varphi\left(E_i\right)$$

□

Definition 1.1.12

- An outer measure φ on a topological space X is called a ***Borel outer measure*** if all Borel sets are φ-measurable.
- A Borel outer measure is ***finite*** if $\varphi(X)$ is finite.
- An outer measure φ on a set X is called ***regular*** if for each $A \subseteq X$ there exists a φ-measurable set $B \supseteq A$ such that $\varphi(B) = \varphi(A)$.
- A ***Borel rugular outer measure*** is a Borel outer measure such that for each $A \subseteq X$, there exists a Borel set B such that $\varphi(B) = \varphi(A)$.
- A ***Radon outer measure*** is a Borel regular outer measure that is finite on compact sets.

Exercises for Outer Measure

Exercise 1.1.13

If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

Let $\varphi_\varepsilon(A) := \varphi(A)$ denote the dependence on ε, and define

$$\psi(A) := \lim_{\varepsilon \to 0} \varphi_\varepsilon(A)$$

What is $\psi(A)$ and what are the correspoding ψ-measurable sets.

Proof. If there are only finite points in A. Let

$$l := \min\{d(x,y) : x, y \in A, x \neq y\}$$

Then for any $x, y \in A$, $x \neq y$. If B_1, B_2 are ball of radius $\frac{l}{2}$ such that $x \in B_1, y \in B_2$. Then $B_1 \cap B_2 = \varnothing$. But each point in A must covered by one of such ball, which shows that $\psi(A) \geq |A|$. Since it is obvious that $\psi(A) \leq |A|$, then $\psi(A) = |A|$.

Now suppose that A is a infinite set. If $\psi(A) < \infty$, say $\psi(A) = n$. Then for any sufficiently small ε, there are balls $B_1, \cdots, B_n$ with radius ε cover A. Take any $n+1$ different points $p_1, \cdots, p_{n+1}$ of A. Let $\varepsilon = \frac{1}{2}\min\left\{d\left(p_i, p_j\right) : i \neq j\right\}$, then $B_1, \cdots, B_n$ cannot cover these $n+1$ points, thus cannot cover A, which is a contradiction. Thus we have $\psi(A) = \infty$.

We find that ψ is just the counting measure on X.Take any $E \subseteq X$,and take any $A \subseteq X$, if A is finite, then $A \cap E, A - E$ are as well, and $|A| = |A \cap E| + |A - E|$, that is $\psi(A) = \psi(A \cap E) + \psi(A - E)$. If A is infinite, then one of the $A \cap E$ and $A - E$ is as well. Then

$$\psi(A) = \infty = \psi(A \cap E) + \psi(A - E)$$

The above shows that E is ψ-measurable. Thus all subset of X are ψ-measurable. $\square$

Exercise 1.1.14

It was shown that $\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1)$ if $E_1 \subseteq E_2$ are φ-measurable with $\varphi(E_1) < \infty$. Prove that this result remains true if E_2 is not assumed to be φ-measurable.

Proof. Since E_1 is measurable,

$$\varphi(E_2) = \varphi(E_2 \cap E_1) + \varphi(E_2 - E_1)$$

but

$$\varphi(E_2 \cap E_1) = \varphi(E_1)$$

which completes the proof. □

1.2 Carathéodory Outer Measure

Definition 1.2.1 ▶ Carathéodory outer measure

An outer measure φ defined on a metric space (X, p) is called a ***Carathéodory outer measure*** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B)$$

whenever A, B are arbitrary subsets of X with $d(A, B) > 0$. Where

$$d(A, B) := \inf\{\rho(a, b) : a \in A, b \in B\}$$

Theorem 1.2.2

If φ is a Carathéodory outer measure on a metric space X, then all closed sets are φ-measurable.

Proof sketch.
用一列集合近似代替 C 是标准的想法. 证明的核心要素有: 1.C 是闭集, 使得按距离向外逼近的行为是良好的. 2. 距离这个属性与 Carathódory 条件是相适应的. 此外, 将命题转化为极限

$$\lim_{i \to \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

是标准的流程. 要证明该极限成立, 则需要一定的技巧. 这里将 $A \setminus C_i$ 和 $A \setminus C$ 相差的中间部分进行细分量化, 将原极限转化为

$$\lim_{i \to \infty} \sum_{j=i}^{\infty} \varphi(T_j)$$

的极限问题, 从而转化为级数的收敛性问题. 将集合隔一个分组, 变成两组两两相隔正距离的集合, 以便套用 Carathéodory 条件.

Proof. Let $C \subseteq X$ be any closed sets. Take $A \subseteq X$, only need to show that

$$\varphi(A) \geq \varphi(A \cap C) + \varphi(A \setminus C)$$

We assume that $\varphi(A) < \infty$. Define

$$C_i := \left\{x : d(x, C) \leq \frac{1}{i}\right\}$$

Since $d(A \setminus C_i, A \cap C) > 0$,

$$\varphi(A) \geq \varphi((A \setminus C_i) \cup (A \cap C)) = \varphi(A \setminus C_i) + \varphi(A \cap C)$$

Only need to show that

$$\lim_{i \to \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

Define

$$T_i := \left\{x \in A : \frac{1}{i+1} < d(x, C) \leq \frac{1}{i}\right\}$$

, then

$$A \setminus C \subseteq (A \setminus C_i) \cup \bigcup_{j=i}^{\infty} T_i$$

Thus

$$\varphi(A \setminus C) \leq \varphi(A \setminus C_i) + \sum_{j=i}^{\infty} \varphi(T_i)$$

Once we have $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, then $\lim_{i \to \infty} \sum_{j=i}^{\infty} \varphi(T_i) = 0$, the conclusion follows from it. To show this, note that $\mathrm{d}\left(T_i, T_j\right) > 0$ whenever $|i - j| \geq 2$. Then it by induction follows from Carathódory condition that

$$\sum_{j=1}^{\infty} \varphi(T_{2i}) \leq \varphi(A) < \infty$$

$$\sum_{j=1}^{\infty} \varphi\left(T_{2j-1}\right) \leq \varphi(A) < \infty$$

Thus $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, which completes the proof. $\square$

Theorem 1.2.3

Suppose $\mathcal{F}$ is a family of subsets of a topological space X that contains all open and all closed subsets of X. Suppose also that $\mathcal{F}$ is closed under countable unions and countable intersections. Then $\mathcal{F}$ contains all Borel sets; that is, $\mathcal{B} \subset \mathcal{F}$.

Note.
可以将包含 $\mathcal{B}$ 描述成对以下三种集合的封闭性: 1. 开集 2. 补集 3. 可数并.
如果我们事先没有对补集的封闭性, 可以通过加强 1.3. 的条件, 使得当我们强制使得补集封闭后, 得到的新集族依旧满足 1.3. 具体地, 我们希望原集族对于开集的补集, 也就是闭集也封闭; 对于可数并的补集, 也就是可数交也封闭.

Proof. Let
$$\mathcal{H} = \mathcal{F} \cap \{A : A^c \in \mathcal{F}\}$$
Obseve that $\mathcal{H}$ contains all closed sets. Moreover, it is easily seen that $\mathcal{H}$ is closed under complementation and countable unions. Thus $\mathcal{H}$ is a σ-algebra that contains the open sets and therefore contains all Borel sets. □

Theorem 1.2.4

If φ is a Carathéodory outer measure on a metric space X, then the Borel sets of X are φ-measurable.

Proof. Since the collection of φ-measurable sets is itself a σ-algebra , it is closed under complements and countable union and countable intersection. Theorem 1.2.2 gives that φ-measurable sets contains closed sets, thus contains open sets as well. Then the conclusion follows from Theorem 1.2.3
. □

Exercises for this section

Exercise 1.2.5 ▸ Smallest σ-algebra containing open sets

Prove that in every topological space X, there exists a smallest σ-algebra that contains all open sets in X. That is, prove that there is a σ-algebra Σ that contains all open sets and has the property that if Σ_1 is another σ-algebra containing all open sets, then $\Sigma \subset \Sigma_1$. In particular, for $X = \mathbb{R}^n$, note that there is a smallest σ-algebra that contains all the closed sets in $\mathbb{R}^n$.

Proof. Let

$$\Sigma := \bigcap_{\mathcal{M}_\alpha \text{ is a } \sigma \text{ algebra containing } X} \mathcal{M}_\alpha$$

Then it is obvious that Σ is closed under complement, countable union, and contains all open sets. Thus Σ is a σ-algebra containing all open sets. □

Exercise 1.2.6

In a topological space X the family of Borel sets, $\mathcal{B}$, is by definition the σ-algebra generated by the closed sets. The method below is another way of describing the Borel sets using transfinite induction. You are to fill in the necessary steps:

1. For an arbitrary family $\mathcal{F}$ of sets, let

$$\mathcal{F}^* = \left\{ \bigcup_{k=1}^{\infty} E_k \ : \ \text{where either } E_i \in \mathcal{F} \text{ or } \tilde{E}_i \in \mathcal{F} \text{ for all } i \in \mathbb{N} \right\}.$$

   Let Ω denote the smallest uncountable ordinal. We will use transfinite induction to define a family $\mathcal{E}_\alpha$ for each $\alpha < \Omega$.

2. Let $\mathcal{E}_0 :=$ all closed sets $:= \mathcal{K}$. Now choose $\alpha < \Omega$ and assume that $\mathcal{E}_\beta$ has been defined for each β such that $0 \le \beta < \alpha$. Define

$$\mathcal{E}_\alpha := \left(\bigcup_{0 \le \beta < \alpha} \mathcal{E}_\beta \right)^*$$

   and define

$$\mathcal{A} := \bigcup_{0 \le \alpha < \Omega} \mathcal{E}_\alpha.$$

3. Show that each $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. Show that $\mathcal{A} \subset \mathcal{B}$.
5. Now show that $\mathcal{A}$ is a σ-algebra to conclude that $\mathcal{A} = \mathcal{B}$.
   (a) Show that $\varnothing, X \in \mathcal{A}$.
   (b) Let $A \in \mathcal{A} \implies A \in \mathcal{E}_\alpha$ for some $\alpha < \Omega$. Show that this implies $\tilde{A} \in \mathcal{E}_\alpha^* \subset \mathcal{E}_\beta$ for every $\beta > \alpha$.
   (c) Conclude that $\tilde{A} \in \mathcal{A}$ and thus conclude that $\mathcal{A}$ is closed under complementation.
   (d) Now let $\{A_k\}$ be a sequence in $\mathcal{A}$. Show that $\bigcup_{k=1}^\infty A_k \in \mathcal{A}$. (Hint: Each A_k is in $\mathcal{A}_{\alpha_k}$ for some $\alpha_k < \Omega$. We know that there is $\beta < \Omega$ such that $\beta > \alpha_k$ for each $k \in \mathbb{N}$.)

Proof. 3. Since $\mathcal{E}_0 \subseteq \mathcal{B}$, only need to show that if $\mathcal{E}_\beta \subseteq \mathcal{B}$ for all $0 \leq \beta < \alpha$, then $\mathcal{E}_\alpha \subseteq \mathcal{B}$. Take any $E \in \mathcal{E}_\alpha$, then there are $E_1, \cdots, E_k, \cdots$, such that $E_i \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, or $E_i^c \in \bigcup_{0 \leq \beta < \alpha} \mathcal{E}_\beta$, and $E = \bigcup_{k=1}^\infty E_k$. We suppose that $E_i \in \mathcal{E}_{\beta_i}$ or $E_i^c \in \mathcal{E}_{\beta_i}$. Since $\mathcal{E}_{\beta_i} \subseteq \mathcal{B}$ for all β_i, and since $\mathcal{B}$ is closed under complement, then $E_i \in \mathcal{B}$. Since $\mathcal{B}$ is closed under countable union, $E = \bigcup_{k=1}^\infty E_k \in \mathcal{B}$. Thus $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. We have shown that $\mathcal{E}_\alpha \subseteq \mathcal{B}$ for all $\alpha < \Omega$, then it is immediate that $A \subseteq \mathcal{B}$.

5. (a) Since $\varnothing \in \mathcal{E}_0$ and $X = (\varnothing)^c \in \mathcal{E}_1 = (\mathcal{E}_0)^*$, then $\varnothing, X \in \mathcal{A}$.

(b) Write

$$\mathcal{E}_\alpha^* = \left\{ \bigcup_{k=1}^\infty E_k \ : \ \text{where either } E_i \in \mathcal{E}_\alpha \text{ or } E_i^c \in \mathcal{E}_\alpha \text{ for all } i \in \mathbb{N} \right\}$$

We take $E_1 = A^c$, then $E_1^c = A \in \mathcal{E}_\alpha$, and take $E_2 = \cdots = \varnothing$.Then $A^c = \bigcup_{k=1}^\infty E_k \in \mathcal{E}_\alpha^*$. If $\beta > \alpha$, then

$$\mathcal{E}_\alpha \subseteq \bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma$$

Let $*$ act on both side, then it follows from the definition of $*$ that

$$(\mathcal{E}_\alpha)^* \subseteq \left(\bigcup_{0 \leq \gamma < \beta} \mathcal{E}_\gamma \right)^* = \mathcal{E}_\beta$$

Thus $A^c \in \mathcal{E}_\alpha^* \subseteq \mathcal{E}_\beta$ for every $\beta > \alpha$.

(c) It is immediate from (b) that $A^c \in \mathcal{E}_\beta \subseteq \mathcal{A}$.

(d) Suppose that $A_k \in \mathcal{E}_{\alpha_k}$ for some $\alpha_k < \Omega$. Let

$$\gamma := \bigcup_{k\in\mathbb{N}} \alpha_k, \quad \beta := \gamma + 1$$

Then $\gamma, \beta < \Omega$, and $\alpha_1, \cdots, \alpha_k < \beta$. Then

$$\bigcup_{k=1}^{\infty} A_k \in \left(\bigcup_{k=1}^{\infty} \mathcal{E}_{\alpha_k}\right)^* \subseteq \mathcal{E}_\beta \subseteq \mathcal{A}$$

□

Exercise 1.2.7 ▶ *

An outer measure φ on a space X is called σ-finite if there exists a countable number of sets A_i with $\varphi(A_i) < \infty$ such that $X \subseteq \bigcup_{i=1}^{\infty} A_i$. Assuming that φ is a σ-finite Borel regular outer measure on a metric space X, prove that $E \subseteq X$ is φ-measurable if and only if there exists an F_σ set $F \subseteq E$ such that $\varphi(E \setminus F) = 0$.

Proof. Let $E_i := E \cap A_i$. Then

$$E = \bigcup_{i=1}^{\infty} E_i, \quad \varphi(E_i) \leq \varphi(A_i) < \infty$$

Once we show that there is a closed set $F_i \subseteq E_i$, such that $\varphi(E_i \setminus F_i) = 0$ for all i, then let $F = \bigcup_{i=1}^{\infty} F_i \in F_\sigma$, it follows that

$$\varphi(E \setminus F) \leq \varphi\left(\bigcup (E \setminus F_i)\right) \leq \varphi\left(\bigcup (E_i \setminus F_i)\right) \leq \sum_i \varphi(E_i \setminus F_i) = 0$$

□

Exercise 1.2.8

Let φ be an outer measure on a space X. Suppose $A \subset X$ is an arbitrary set with $\varphi(A) < \infty$ such that there exists a φ-measurable set $E \supset A$ with $\varphi(E) = \varphi(A)$. Prove that $\varphi(A \cap B) = \varphi(E \cap B)$ for every φ-measurable set

B.

Proof.

$$\varphi(A) = \varphi(E) = \varphi(E \cap B) + \varphi(E \setminus B)$$

$$\begin{aligned}\varphi(A \cap B) \geq \varphi(A) - \varphi(A \setminus B) &= \varphi(E) - \varphi(A \setminus B) \\ &= \varphi(E \cap B) + \varphi(E \setminus B) - \varphi(A \setminus B) \\ &\geq \varphi(E \cap B)\end{aligned}$$

Since $A \cap B \subseteq E \cap B$, $\varphi(A \cap B) \leq \varphi(E \cap B)$. Finally, we have $\varphi(A \cap B) = \varphi(E \cap B)$ □

Exercise 1.2.9

In $\mathbb{R}^2$, find two disjoint closed sets A and B such that $\mathbf{d}(A, B) = 0$. Show that this is not possible if one of the sets is compact.

Proof.

1. Let A be the graph of $y = \frac{1}{x}$. Let $B = \{(x, y) : y = 0\}$. Then $\left\{\left(n, \frac{1}{n}\right) : n \in \mathbb{N}\right\} \subseteq A$, $\{(n, 0) : n \in \mathbb{N}\} \subseteq B$, then

$$\mathbf{d}(A, B) \leq \inf\left\{d\left(\left(n, \frac{1}{n}\right), (n, 0)\right) : n \in \mathbb{N}\right\} = \lim_{n\to\infty} \frac{1}{n} = 0$$

2. If A is compact, B is closed. Since $\mathbf{d}(A, B) = 0$, there are convergent sequences $\{a_n\} \subseteq A$ and $\{b_n\} \subseteq B$ such that $\lim_{n\to\infty} d(a_n, b_n) = 0$. Since A is bounded and closed, there is a cluster point $a \in A$ of $\{a_n\}$. Suppose that $\lim_{k\to\infty} a_{n_k} = a$, then $\lim_{k\to\infty} b_{n_k} = a \in B$. $A \cap B \neq \varnothing$, which is a contradiction.

□

Exercise 1.2.10

Let φ be an outer Carathéodory measure on $\mathbb{R}$ and let $f(x) := \varphi(I_x)$, where I_x is an open interval of fixed length centered at x. Prove that f is lower semicontinuous. What can you say about f if I_x is taken as a closed interval? Prove the analogous result in $\mathbb{R}^n$; that is, let $f(x) := \varphi(B(x, a))$, where $B(x, a)$ is the open ball with fixed radius a centered at x.

Exercise 1.2.11

In a metric space X, prove that $\operatorname{dist}(A, B) = d(\bar{A}, \bar{B})$ for arbitrary sets $A, B \in X$.

Exercise 1.2.12

Let A be a non-Borel subset of $\mathbb{R}^n$ and define for each subset E,

$$\varphi(E) = \begin{cases} 0 & \text{if } E \subset A, \\ \infty & \text{if } E \setminus A \neq \varnothing. \end{cases}$$

Prove that φ is an outer measure that is not Borel regular.

Exercise 1.2.13

Let $\mathcal{M}$ denote the class of φ-measurable sets of an outer measure φ defined on a set X. If $\varphi(X) < \infty$, prove that the family

$$\mathcal{F} := \{A \in \mathcal{M} : \varphi(A) > 0\}$$

is at most countable.

1.3 Lebesgue Measure

Theorem 1.3.1

Suppose each edge $I_k = [a_k, b_k]$ of an n-dimensional interval I is partitioned into α_k subintervals. The products of these intervals produce a partition of I into $\beta := \alpha_1 \cdot \alpha_2 \cdots \alpha_n$ subintervals I_i and

$$v(I) = \sum_{i=1}^{\beta} v(I_i).$$

Theorem 1.3.2

For each interval I and each $\epsilon > 0$, there exists an interval J whose interior contains I and

$$v(J) < v(I) + \epsilon.$$

Definition 1.3.3 ▸ Lebesgue outer measure

The ***Lebesgue outer measure*** of an arbitrary set $E \subseteq \mathbb{R}^n$, denoted by $\lambda^*(E)$, is defined by

$$\lambda^*(E) := \inf\left\{\sum_{k=1}^{\infty} v(I_k)\right\},$$

where the infimum is taken over all countable collections of closed intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k.$$

Remark.

- It may be necessary at times to emphasize the dimension of the Euclidean space in which Lebesgue outer measure if defined. When clarification is needed, we will write $\lambda_n^*(E)$ in plce place of $\lambda^*(E)$.
- We have not prove that ***Lebesgue outer measure*** is actually a outer measure till now.

Theorem 1.3.4

For a closed interval $I \subseteq \mathbb{R}^n$, $\lambda^*(I) = v(I)$.

Proof. The inequality $\lambda^*(I) \leq v(I)$ holds, since I itself givs a element of $\left\{\sum_{k=1}^{\infty} v(I_k)\right\}$ in the definition of $\lambda^*(I)$.

To prove the opposite inequality, choose $\varepsilon > 0$, and a sequence of closed intervals $I_1, I_2, \cdots$ such that

$$I \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \leq \lambda^*(E) + \varepsilon$$

We can choose closed intervals J_k such that

$$v(J_k) \leq v(I_k) + \frac{\varepsilon}{2^k}, \quad I_k \subseteq J_k^\circ$$

Let $\mathcal{F} := \left\{J_k^\circ : k \in \mathbb{N}\right\}$. Then $\mathcal{F}$ is an open cover of I, since I is compact, it has a lebesgue number η. There is a finitely many partition $K_1, \cdots, K_m$ of I such that each with diameter less than η. Then K_i is contained in some element of $\mathcal{F}$, we say J_{k_i} althoung two K_i may be contained in a same J_{k_i}. Let N_m be the smallest number of J_{k_i} containing K_i, then

$$v(I) = \sum_{i=1}^{m} v(K_i) \le \sum_{i=1}^{N_m} v\left(J_{k_i}\right) \le \sum_{k=1}^{\infty} v(J_k) \le \lambda^*(E) + 2\varepsilon$$

Since ε is arbitrary, $v(I) \le \lambda^*(E)$. $\square$

Theorem 1.3.5

Lebesgue outermeasure, λ^*, defined on $\mathbb{R}^n$ is a Carathéorody outer measure.

Proof. We first verify that λ^* is an outer measure. We only show the subadditivity of λ^*, the other conditions are obvious. Let $\{A_i\}$ be a countable collection of arbitrary subsets of $\mathbb{R}^n$. Denote $A := \bigcup_{i=1}^{\infty} A_i$. There is a countable family of closed intervals $\left\{I_j^{(i)}\right\}_{j\in\mathbb{N}}$ such that

$$A_i \subseteq \bigcup_{j=1}^{\infty} I_j^{(i)}, \quad \sum_{j=1}^{\infty} v\left(I_j^{(i)}\right) < \lambda^*(A_i) + \frac{\varepsilon}{2^i}$$

Collection of closed interval $\left\{I_j^{(i)}\right\}_{i,j\in\mathbb{N}}$ can be aranged to a sequence of closed invervals, such that

$$A \subseteq \bigcup_{i,j\in\mathbb{N}} I_j^{(i)}$$

Thus

$$\lambda^*(A) \le \sum_{i,j} v\left(I_j^{(i)}\right) \le \sum_{i=1}^{\infty} \lambda^*(A_i) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, the countable subadditivity of λ^* is established.

To show that λ^* is Carathéorody outer measure. We take any $\varepsilon > 0$, and take a countable conllection of closed intervals $I_1, I_2, \cdots$ such that

$$A \cup B \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \subseteq \lambda^*(A \cup B).$$

We can subdividing each I_k into smaller invervals if necessary, we may assume that each interval I_k which diameter less that $\mathbf{d}(A, B)$. Then $\{I_k\}$ can be seperated into two subfamiles $\{I_k'\}$ and $\{I_k''\}$. The members in the former have empty intersection with B , and that in the later have empty intersection with A. $\{I_k\} = \{I_k'\} \sqcup \{I_k''\}$. Thus

$$A \subseteq \bigcup_{k=1}^{\infty} \{I_k'\}, \quad B \subseteq \bigcup_{k=1}^{\infty} \{I_k''\}$$

We have

$$\begin{aligned} \lambda^*(A) + \lambda^*(B) &\leq \sum_{k=1}^{\infty} v(I_k') + \sum_{k=1}^{\infty} v(I_k'') \\ &= \sum_{k=1}^{\infty} v(I_k) \\ &< \lambda^*(A \cup B) + \varepsilon \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) + \lambda^*(B) \leq \lambda^*(A \cup B)$$

□

Corollary 1.3.6

All Borel sets in $\mathbb{R}^n$ are Lebesgue measurable. In particular, each open set and each closed set are Lebesgue measurable.

Definition 1.3.7 ▶ Lebesgue Measure

Donote by λ the set function obtained by restricting λ^* to the family of Lebesgue measurable sets. λ is called ***Lebesgue measure***.

Corollary 1.3.8

Let J be an opoen interval in $\mathbb{R}^n$. Then

$$\lambda(J) = \mathrm{vol}(J)$$

Proof. There is a sequence of closed intervals I_k such taht $J = \bigcup_{k=1}^{\infty} I_k$, then

$$\lambda(J) = \lim_{k\to\infty} \lambda(I_k) = \lim_{k\to\infty} \operatorname{vol}(I_k) = \operatorname{vol}(J)$$

□